MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY (BICTU)

END OF SEMESTER EXAMINATION

ICT 4703 : NEWORK ADMINISTRATION AND INFORMATION SECURITY

TIME ALLOWED: 3 HOURS **DATE : 11th SEPTEMBER , 2017**

INSTRUCTIONS

i. This paper contains **FIVE**pages. Please check.
ii. There are **TEN**in this paper. Please check.
iii. Answer **ALL** questions.
iv. Marks for each question are indicated.

SECTION A: MULTIPLE CHOICE QUESTIONS [10 Marks]

1. Which of the following correctly describes the characteristics of switched and routed data flows?

   A. Switches create a single collision domain and a single broadcast domain,Routers provide separate collision domains.
   B. Switches create separate collision domains and separate broadcast domains, Routers provide separate collision domains and separate broadcast domain as well.
   C. Switches create separate collision domains and separate broadcast domains, Routers provide separate collision domains.
   D. Switches create a single collision domain and separate broadcast domains, Routers provide separate broadcast domain as well.
   E. Switches create separate collision domains and a single broadcast domain, Routers provide separate broadcast domain.

2. Which of the following is a true statement about the function of a bridge?
   A. A bridge creates or maintains a table of the available routes and their conditions.
   B. A bridge passes packets outside of its network segment if the IP address cannot be found on its table.
   C. A bridge sends the frame towards the destination host after it looks up the frame's destination in its address table.
   D. A bridge creates or maintains a table of the IP addresses of the hosts connected to its network segment.

3. Which of the following symbols in the router prompt indicates that you are in privileged mode?
   A. %
   B. #
   C. Privileged
   D. >
   E. ^

4. You are configuring a cisco switch. As you try to enter a command, you keep getting the % incomplete error response. How can you get help on the command that you are trying to enter?
   A. Reenter the command followed by a question mark to view keywords.
   B. Type " Help" followed by the command to see command parameters.
   C. Use the up arrow to review the prompt before the error.
   D. Type "?" to display all the console commands.
   E. Type history to review the prompt before the error.

5. How do you configure the line console password to "cisco"?
   A. Line console 0, Login, Password Cisco.
   B. Line console 0 4, Login, Password cisco.
   C. Line console, Login, Password cisco.
   D. Line console 0, Login, Password cisco.

6. You are required to back up your IOS to a network location. Which of the following command will accomplish this?
   A. copytftp flash.
   B. copy flash ftp.
   C. copy ftp flash.
   D. copy flash tftp.

7. Which layer provides segmentation of data and reassembling at the other end?
   A. Session.
   B. Transport.
   C. Datalink.
   D. Network.

8. What command will enable an encrypted password which cannot be seen in clear text in the running config?
   A. Enable password.
   B. Enable encrypted password.
   C. Service password encrypted.
   D. Enable secret password.

9. Which command will enable a network on an interface?
   A. Ip network 10.1.2.0.
   B. Network 10.1.0.0.
   C. Network Interface 10.1.0.0 255.255.0.0.
   D. Network 10.0.0.0.

10. You are given an ipadress 132.15.136.2/18 what subnet is the host ipon ?
   A. 132.15.136.0.
   B. 132.15.128.0.
   C. 132.15.192.0.
   D. 132.15.17.0.

SECTION B : STRUCTURED QUESTIONS **[40 Marks]**

1. Remote access is often used by system administrators in large organisations to remotely control the computers on the organisation's local area network. Explain how malicious remote access administration tools (RATs) can be used. **[10 Marks]**

2. You work in a multi-branched company. You have been given a task to assist management in trimming unnecessary ICT-related costs (Travel expenses, phone) as much as possible. As a Network Administrator with sound and updated knowledge, what would you do? **[10 Marks]**

3. Explain how Cryptography achieves the security goals. **[10 Marks]**

4. Suppose Alice wants to send "Sweetheart" to Bob using vigenere polyalphabetic cipher using key=ENOCH. Calculate the cipher text that Bob will receive. **[10 Marks]**

The Vigenère Square

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
2	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
3	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
4	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
5	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
6	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
7	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
8	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
9	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
10	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
11	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
12	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
13	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
14	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
15	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
16	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
17	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
18	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
19	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
20	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
21	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
22	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
23	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
24	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
25	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
26	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z

SECTION C : NETWORK DESIGN **[50 Marks]**

TechNix Malawi Ltd. Network requirements

Youhavebeengiventhenetworkaddress10.51.0.0/24to configureandprovidetheIP addressingforthenetworkshowninFigure 2.Thenetworkhasthefollowingaddressing requirements:

- TheLILONGWELAN1willrequire20hostIPaddresses.
- TheLILONGWELAN2willrequire 15 hostIPaddresses.
- TheMZUZULAN1willrequire30hostIPaddresses.
- TheMZUZULAN2willrequire10hostIPaddresses.
- TheHQ-BTLANwillrequire50hostIPaddresses.
- Mozambique LAN will require 49 Addresses.
- Tanzania LAN will require 27 Addresses.
- The links between routers will require an IP Address at each end.

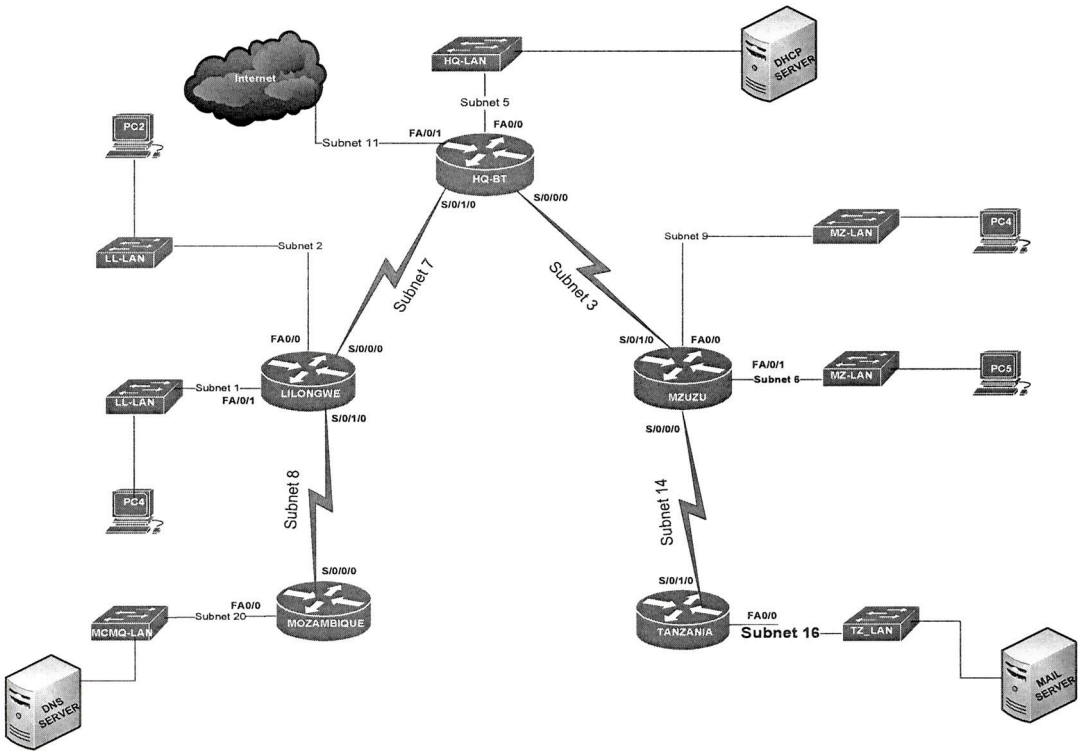


Figure 2. TechNix Malawi Ltd. Network Infrastructure

a) Perform basic configuration on the routers and switches including MOTD. **[10 Marks]**

b) Assigntheappropriateaddressestothedeviceinterfaces using the following instructions: **[15 Marks]**

   i. Assign the first valid address to the serial interfaces s0/0/0 and the second valid address to S0/1/0 on each subnet between routers.

   ii. Assign the last valid address to all PCs.

   iii. Assign the third valid address to all the servers.

c) Implement Static routing on the network. **[10 Marks]**

d) Cofigure the default route on each router **[5 Marks]**

e) What command would enable OSPF? **[5 Marks]**

f) How would you backup config on a remote tftp server with an address 10.168.0.100? **[5 Marks]**

END OF QUESTION PAPER